WOWII: A counterexample to conjecture 91

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1 Conjecture

If G is a simple connected graph, then $b(G) \leq 1 + f(G) \cdot \frac{\lceil l_{avg}(v) \rceil}{2}$

Conjecture reference.

Notations

- $b(G)$: The maximum number of vertices of an induced bipartite subgraph of the graph.
- $f(G)$: The number of vertices of a largest induced forest of the graph.
- $l(v)$: The independence number of the subgraph induced by the neighbors of vertex v.
- l_{avg}: The average of the $l(v)$ values for all vertices of the graph.

2 Results

2.1 Graph identification

The graph is defined by the following `graph6` string:

```
[?@jcAW?G@_F?@?@_?w?NQ????G??W??[??N?@{C_?????@???B???B???@W???N
```

2.2 Calculated properties

Property	Value
Number of vertices n	28
Number of edges m	57
Bipartite number $b(G)$	18
Forest number $f(G)$	16
l_{avg}	1.9642857

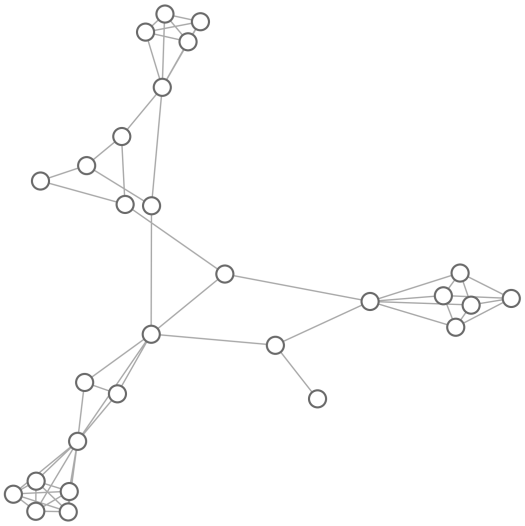


Figure 1: Counterexample

3 Verification of the inequality

The values of $l(v)$ for each vertex v are listed below:

$$\{3, 3, 2, 3, 3, 3, 3, 1, 4, 1, 1, 2, 1, 1, 1, 1, 3, 1, 1, 1, 1, 1, 4, 2, 2, 2, 2, 2\}$$

$l_{avg} = \frac{55}{28} = 1.964$

Let us calculate the right-hand side of the conjecture:

$$\text{RHS} = 1 + \frac{f(G) \cdot \lceil l_{\text{avg}} \rceil}{2} = 1 + \frac{16 \cdot \lceil 1.964 \rceil}{2} = 1 + \frac{16 \cdot 2}{2} = 17.$$

The left-hand side is:

$$\text{LHS} = b(G) = 18.$$

Verification:

$$18 > 17 \implies \textbf{FALSE}$$

4 Conclusion

The conjecture states that $b(G) \leq RHS$. However, we observe:

$$b(G) = 18 \quad \text{and} \quad \text{RHS} = 17$$

$$18 > 17$$

The inequality is not satisfied. **This graph is therefore a valid counterexample.**